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Inelastic behaviour of ceramic-matrix composites

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Abstract

A methodology for the formulation and identification of the constitutive laws of ceramic-matrix composites is summarised. It relies on an anisotropic damage evaluation that accurately separates the effects of the various damage mechanisms on the non-linear behaviour. A mixed approach takes into account the basic strain and damage mechanisms by using a homogenisation method that provides the relationship between the mechanical response and the intensity of damage in the individual modes. That leads to a non-arbitrary choice of internal variables in the macroscopic constitutive relationships. A successive process of prediction/experimental-data confrontation allows the optimal determination of the evolution laws of those internal variables. This methodology is illustrated on various behaviours of various CMCs; several crack arrays, tilted cracks, tensile test, cyclic loading, off-axis solicitation, in 1D SiC–SiC, 2D C–SiC, 2D C/C–SiC ceramic-matrix composites. Predictions of the three-dimensional changes in elasticity and of the inelastic strains are shown to compare favourably with experimental data measured with an ultrasonic method. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: A. Ceramic-matrix composites; B. Modelling; Non-linear behaviour; Matrix cracking; C. Anisotropy; Damage mechanics; Elastic properties; D. Ultrasonics

1. Introduction

The macroscopic mechanical behaviour of ceramic-matrix composites is strongly influenced by the onset and the development of microcracks [1,2]. The behaviour of CMCs is the result of the combination of two main damage mechanisms [3]: matrix microcracking normal to the tensile axis, deflection of these cracks at the fibre-matrix interface if the interface is weak enough (Fig. 1). The matrix microcracking induces a loss of stiffness. Mode II cracking prevents the composite from failing too early because the fibre-matrix debonding leads to a fibre-matrix sliding with friction depending on the nature of the interface [4,5]. Other mechanisms increase failure energy absorption like fibre pull-out and out of matrix crack-plane fibre fracture [6].The combination of these mechanisms leads to a highly non-linear behaviour (Fig. 2).

In most CMCs, debonding is accompanied by frictional sliding which turns loading/unloading cycles into hysteresis loops (Fig. 2). The extent of the inelastic strains and the area of the hysteresis loops result from both an intense interfacial debonding and fibre-matrix frictional sliding. Limited hysteresis loops account for negligible frictional sliding while debonding, on the other hand, can be broadly present.

To formulate the constitutive laws of such materials, it is important to separate the effects of initiation and growth of microcracks from the effects due to the presence of cracks. The various damage mechanisms induced by mechanical loading and their influence on the tensile behaviour were determined and analysed by comparing experimental variations of the components of the stiffness tensor obtained from ultrasonic measurements and prediction of effective stiffness properties of medium permeated by cracks. The relationships between the effective stiffness tensor and the intensity of damage in individual modes, provide coherent and comprehensive physical explanations for the observed experimental phenomenology. Various scales are considered: the micro-scale at which the damage mechanisms are described and the macro-scale where the volume element is large enough to consider that the discrete damage mechanisms are well represented by a mean leading to continuous variables. The major point of the methodology lies in the non-arbitrariness of the choice of the internal variables with a concrete physical meaning which reflect the underlying processes on the microscale, as example, the cracks density or the crack opening displacement.

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2. Experimental

An ultrasonic device [7] and an extensometer used simultaneously provide a useful method for carrying out a strain partition under load [8]. Applied to CMCs, the contribution of the damage mechanisms to their highly non linear behaviour can be evaluated.

2.1. Ultrasonic method

The ultrasonic device consists in an immersion tank associated to a tensile machine [7]. It allows to study the complete stiffness tensor variation under load thus it is possible to know which coefficients are affected during the damage process.

Wave speed measurements are performed by using ultrasonic pulses which are refracted through the sample immersed in water. The measurement of the phase velocity of the pulses is done by a signal processing method using Hilbert transform [9].

To fully describe the elastic behaviour of an orthotropic material, the nine elastic constants C_{ij} are identified by measuring the phase velocities of bulk waves transiting in two accessible principal planes [planes (1, 2) and (1, 3), Fig. 3] and in a non principal plane [plane (1, 45°] described by the bisectrix of axis 2 and 3) of the sample[10]. The identification in plane (1, 2) gives four elastic constants; C_{11}, C_{22}, C_{66} and C_{12} and three others are obtained in plane (1, 3): C_{33}, C_{55} and C_{13}. The two remaining coefficients C_{23} and C_{44} are identified by propagation in the non-principal plane (1, 45°). The confidence interval associated to each identified constant is then estimated by a statistical analysis [11].

2.2. Under load strain partition

Unloading–reloading cycles are usually carried out to measure the plastic strains. In the plasticity of metals, dislocations are the main flaws. They lead to relative discompositions that remain stable after complete unloading, and which found expression, from a macroscopic point of view, in permanent strains. The introduction of the term "inelastic" is lead by a possible part-recovery of the strains beyond the yield point. It is all the more justified for brittle matrix composites. The matrix microcracking does not modify a pre-existing elastoplastic behaviour but creates residual strains that could not exist otherwise. The inelastic strains in CMCs are the macroscopic indication of the transverse crack opening displacement (COD) due to interfacial fibre-matrix sliding [12]. This sliding is subordinate to the direction of the load and thus partly reversible. A

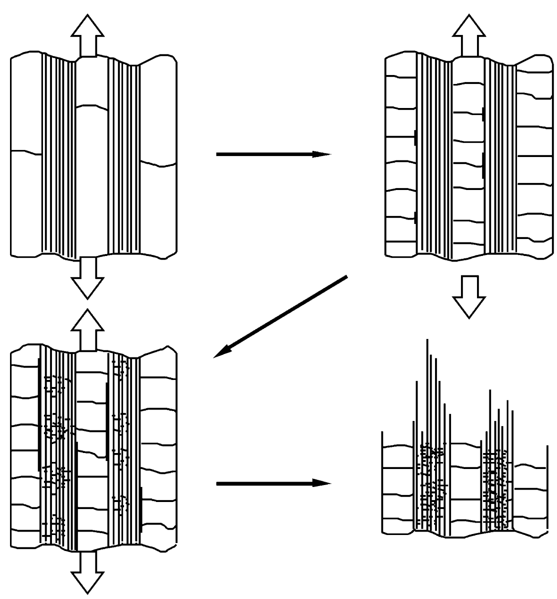

Fig. 1. Damage mechanisms in CMCs.

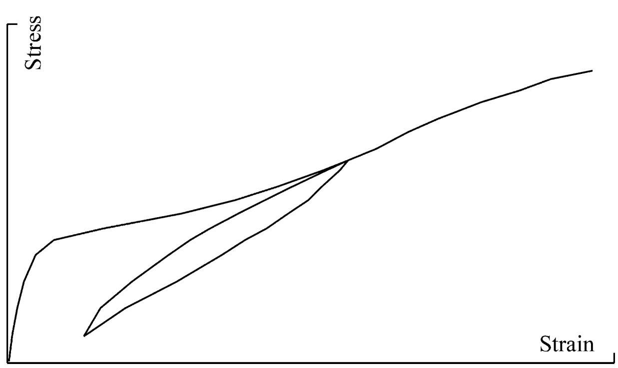

Fig. 2. Consequence of matrix cracking and fibre-matrix sliding on the stress/strain behaviour.

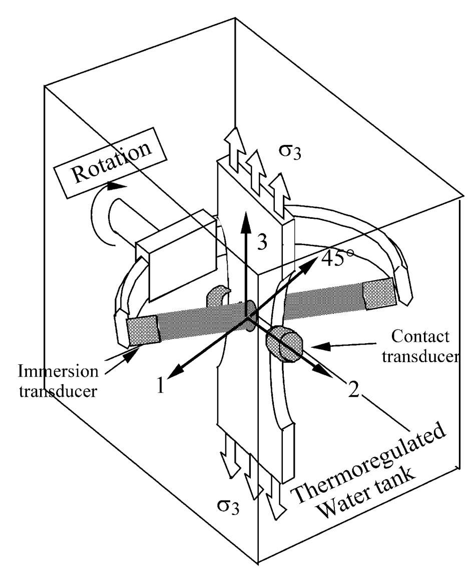

Fig. 3. Sample instrumented for strain partition under load in the ultrasonic device used to investigate the stress-induced development of the damage of CMCs.

decrease of the stress can create a sliding opposite to the one created by an increase in stress. Therefore, the strain measured when the specimen is completely unloaded only represents the permanent part of the inelastic strain. The inelastic part of the strain is underestimated by the classic measurements using unloading–reloading cycles.

The matrix microcracking induces an important loss of the elastic modulus. It is usually [13] measured using the slopes of unloading-reloading cycles. This measurement soon becomes inaccurate as the fibre-matrix sliding is accompanied by a frictional effect in the crack wake. Unload-reload curves turns into hysteresis [1] (Fig. 4). The hysteresis tangent is subordinate to both the elastic modulus drop and the reversible inelastic strains. The apparent strain soon includes both the elastic part and the inelastic part, as the sliding threshold is equal or very close to the stress reached before unloading (or to the minimal stress reached before reloading). In another way, measuring the elastic modulus from the slopes of unloading–reloading cycles can only be correct if the sliding occurs for a stress that is different enough from the one reached at the loading or unloading point. Therefore, the choice of the origin of a hysteresis tangent is a particularly hazardous task. It is all the more hazardous when unloading–reloading cycles induce some closing–opening effects on the microcracks as pointed out by the changes of slope in the hysteresis [14].

Consequently, the usual identification underestimates the inelastic strains and overestimates the elastic modulus drop because the reversible part of the inelastic strains can not be separated from the elastic strain. To separate and to identify the two damage mechanisms responsible for the non-linear behaviour of CMCs, it is necessary to make a strain partition under load.

The complete determination of the compliance tensor together with its variation during a tensile test allows to know the elastic part of the material behaviour. The elastic strains are obtained from the generalised Hooke's law:

$$\varepsilon^{e} = \left(C_{ij}\right)^{-1}\sigma_{j} = S_{ij}\cdot\sigma_{j}. \tag{1}$$

When the total strain is measured simultaneously with the ultrasonic evaluation, it becomes possible to know precisely which part of the global strain is either elastic or inelastic. As the extensometer indicates the total strain (Fig. 4) and since the variation of the elastic strain is given by Eq. (1) with the variation of C_{ij}, the inelastic strain is then simply obtained [8]:

$$\varepsilon^{in} = \varepsilon - \varepsilon^{e}. \tag{2}$$

3. Constitutive laws

The failure mechanisms favour the generation of microcracks oriented normal to the tensile stress. The microcracks tend to propagate rapidly inside the brittle matrix across the entire width of fibre spacing. This width is thus quasi-immediately the length of each crack. The damage process is then the multiplication of matrix cracks that propagate perpendicular to the tensile direction (Fig. 5). These cracks appear to be homogeneously distributed. The number of cracks does not change continuously but may increase step by step. However, this increment is very small and the density of the matrix microcracks can be assumed to be continuous. The chosen damage variable is the crack density β associated with the parameters needed to define the geometry of a cracks system with a given orientation. It measures the average distance between cracks.

The inelastic strain along the tensile axis represents a large part of the total strain in CMCs. It is now commonly recognised that they have their source in the sum of the

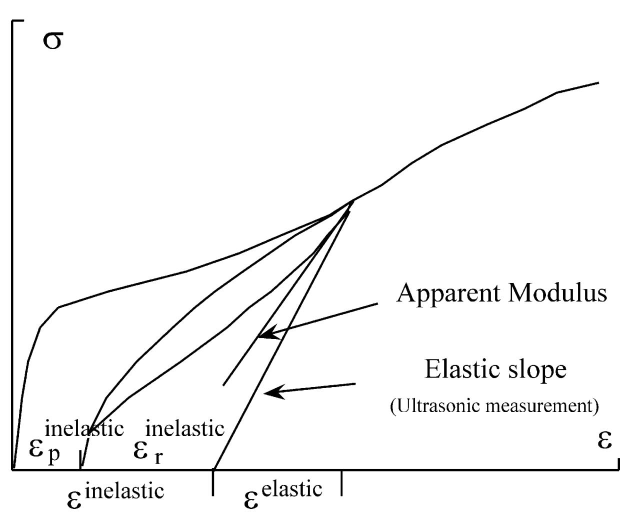

Fig. 4. Strain partition under load.

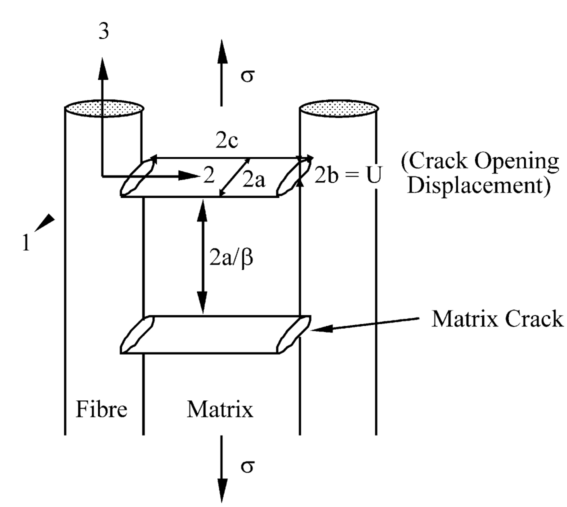

Fig. 5. Unit cell of a cracked body.

crack opening displacements of the transverse crack system, U (Fig. 6) [12]. The distance between two cracks is related to the crack density. So, in an extensometer length L, the number n of cracks, $2a$ deep and $2b$ thick, is:

$$n = L\frac{\beta}{2a}. \tag{3}$$

The relationship between the inelastic strain and the crack opening displacement is then [15]:

$$\varepsilon^{\text{in}} = \frac{\Delta L^{\text{in}}}{L} = \frac{n \cdot 2U}{L} = \frac{n \cdot 2b}{L} = \delta \cdot \beta. \tag{4}$$

Thus, the inelastic strain is simply the density of transverse matrix microcracks multiplied by their aspect ratio $\delta = b/c$.

The extension of the cracks is limited by the waviness of the bundles and experimental observation of inelastic strains implies that crack opening displacement is not negligible. Therefore, it is necessary to consider 3D-defined cracks in order to evaluate the effective stiffness tensor of the damaged material [16]. The cracks are modelling by ellipsoidal voids. Their volume concentration is defined through a unit cell; it represents the largest volume of material containing a single crack.

By using a homogenisation method that provides the relationship between the effective stiffness tensor and the intensity of damage in the individual modes, it is possible to relate the micro- and macro-level damage measurement. The cracked material is substituted by an equivalent homogeneous medium. Effects of damage are then described by the changes of the effective properties of the equivalent medium.

Cracks are consider as an ellipsoidal inclusion:

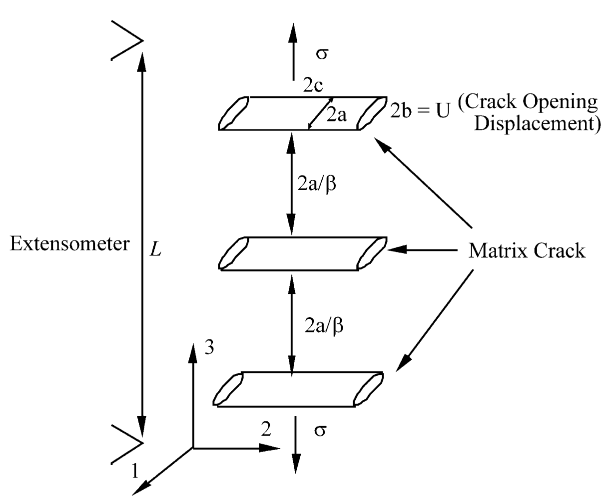

Fig. 6. Inelastic strains; a macroscopic consequence of the microscopic crack opening displacement.

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{c^2} + \frac{x_3^2}{b^2} \leqslant 1, \tag{5}$$

with stiffness C_r and compliance S_r, which is embedded in an infinite homogeneous solid whose stiffness and compliance tensors are, respectively, C and S. The material is loaded by uniform stress, $\bar{\sigma}$, or subjected to uniform strain, $\bar{\varepsilon}$, at infinity. Let the stress and strain fields in the inclusion be σ_r and ϵ_r, respectively, so that:

$$\sigma_r = C_r\varepsilon_r, \ \ \varepsilon_r = S_r\sigma_r. \tag{6}$$

It is well known that the elastic field in the ellipsoidal inclusion is uniform [17] and can be evaluated as

$$\sigma_r = B_r\bar{\sigma} \ , \varepsilon_r = A_r\bar{\varepsilon} \tag{7}$$

A_r and B_r are the crack localisation tensors of the r inclusion:

$$B_r = [I + Q(S_r - S)]^{-1}, \quad A_r = [I - P(C_r - C)]^{-1}. \tag{8}$$

The solution of this problem requires the determination of the tensor Q defined by:

$$Q = C - CPC \tag{9}$$

and the determination of the tensor P whose components are given by [18]:

$$P_{ijkl} = \frac{abc}{4\pi}\int_{\Omega} \frac{D_{ijkl}(\omega_n)}{\left(a^2\omega_1^2 + c^2\omega_2^2 + b^2\omega_3^2\right)^{3/2}} d\Omega \tag{10}$$

where Ω is the surface of the unit sphere centred at the origin of $(\omega_1, \omega_2, \omega_3)$ space. The fourth order tensor D is defined by:

$$D_{ijkl} = \omega_l\omega_j g_{ik} \quad \text{with} \quad g_{ik} = \left[C_{mnpq}\omega_n\omega_q\right]_{ik}^{-1}. \tag{11}$$

Turning now to the basic equations for composites, we note that in order for the concept of overall moduli to be meaningful, it is necessary to consider macroscopically uniform loading [19]. In such a case, the applied stress is equal to the average stress, σ, and the phase average stresses and strains are related to the overall averages through

$$\bar{\sigma}_r = B_r\bar{\sigma} \quad \text{and} \quad \bar{\varepsilon}_r = A_r\bar{\varepsilon}. \tag{12}$$

Let c_r denote the volume concentration of the rth phase. Since

$$\sum_r c_r = 1, \ \bar{\sigma} = \sum_r c_r\bar{\sigma}_r, \ \bar{\varepsilon} = \sum_r c_r\bar{\varepsilon}_r, \tag{13}$$

it follows [20] that the overall stiffness C and compliance S are given by:

$$C_{\mathrm{eff}} = \sum_{\mathrm{r}} c_{\mathrm{r}} C_{\mathrm{r}} A_{\mathrm{r}} \quad \text{and} \quad S_{\mathrm{eff}} = \sum_{\mathrm{r}} c_{\mathrm{r}} S_{\mathrm{r}} B_{\mathrm{r}}. \tag{14}$$

The effective medium is considered as a two-phase medium [21]. Let phase 1 be the uncracked fibre reinforced composite material. Phase 2 is a set of ellipsoids consisting of voids. The overall stiffness tensor for the two-phase medium follows from Eq. (14):

$$C_{\mathrm{eff}} = C - c_{\mathrm{f}} C(C - C_{\mathrm{f}}) A_{\mathrm{f}}, \quad S_{\mathrm{eff}} = S + c_{\mathrm{f}}(S - S_{\mathrm{f}}) B_{\mathrm{f}} \tag{15}$$

where f index is for ellipsoidal voids. The crack localisation tensors are

$$A_{\mathrm{f}} = (I - PC)^{-1} \quad \text{and} \quad B_{\mathrm{f}} = -Q^{-1}, \tag{16}$$

and the effective elasticity tensor is given by

$$C_{\mathrm{eff}} = C - c_{\mathrm{f}} C(I - PC)^{-1} \text{ and } S_{\mathrm{eff}} = S + c_{\mathrm{f}} Q^{-1} \tag{17}$$

with c_{f} the volume concentration of the cracks:

$$c_{\mathrm{f}} = \frac{4\pi}{3} \frac{abc}{2x_1 2x_2 2x_3} \tag{18}$$

where $2x_{\mathrm{i}}$ are the average distances between two cracks in the i direction. They give the unit cell of the cracked material (Fig. 7).

P and Q tensors appearing in Eq. (9) depend upon the shape of the considered inclusion and the stiffness of the effective medium C. That leads to the self-consistent scheme. Here, we replace C with C_0, the stiffness tensor of the uncracked material, in Eqs. (9), (16) and (17). This method, similar to the Mori–Tanaka method [22], requires less calculations and gives a good approximation of the effective stiffness tensor for reasonable volume concentration of cracks [16].

Eq. (17) is the equation used to evaluate the effective stiffness tensor for an anisotropic medium permeated by ellipsoidal cracks. It requires the determination of the tensor Q and P by a numerical evaluation of Eq. (10) [16]. P, the symmetrized derivative of the Green's tensor,

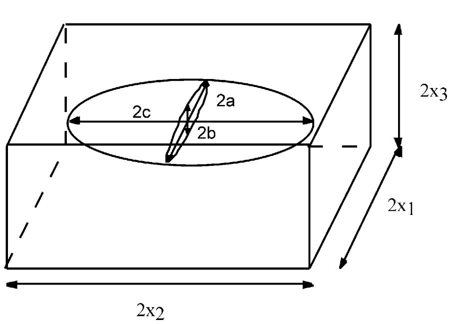

Fig. 7. Unit cell of a cracked body.

depends on the both the geometry of the crack and the elastic properties of material surrounded the crack: here, the initial non cracked material.

4. Several crack arrays

4.1. 1D SiC–SiC

The 3D representation of the cracks was applied to a multiple arrays microcracking in a unidirectional SiC–SiC.

Micrographs (Fig. 8) and experimental stiffness tensor changes [23] (Fig. 12) allow us to identified three arrays of microcracks: a transverse microcracks, which is stopped and deviated in longitudinal cracks at the fibre matrix interface. Those longitudinal arrays are growing cracks along the interface (Fig. 9).

A successive process of prediction experimental data confrontation allows the optimal determination [24] of the fixed sizes of the multiplication of the transverse cracks and of evolution laws of the cracks density, of the cracks thickness aspect ratio (Fig. 10) and of the semi-axes of the growing longitudinal cracks (Fig. 11).

Fig. 12 plots the changes of the nine stiffnesses identified from the phase velocities as a function of tensile stress for the 1D SiC–SiC. There is a good agreement

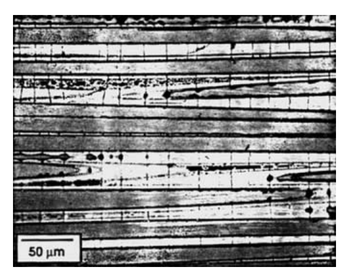

Fig. 8. Micrograph of the transverse cracks in the 1D SiC–SiC.

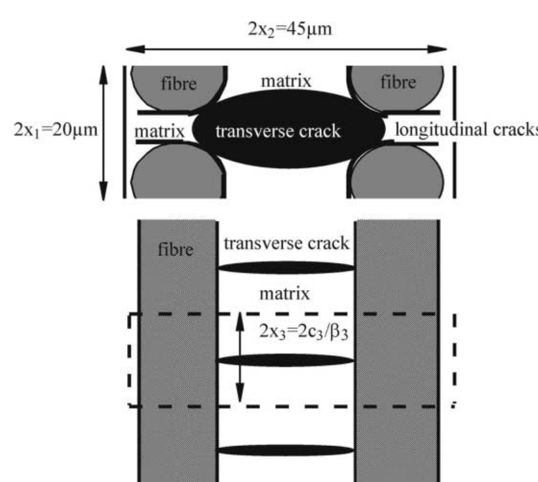

Fig. 9. Unit cell of the cracked 1D SiC–SiC.

between experimental values and the prediction, Eq. (17), plotted in straight lines [24].

4.2. 2D C–SiC

Let us consider a woven 2D C–SiC. Micrographs [25] (Fig. 13) and ultrasonic evaluation of the stiffness tensor changes (Fig. 18) allow us to identified three arrays of microcracks. The damage starts with a transverse cracking localised in the matrix mantle that surrounds the longitudinal bundle (Fig. 14). The waviness of the bundles stops the matrix cracking, which is deviated in mode II by the transverse bundles. Then, longitudinal cracking arrays normal to direction 2 or 1 are created (Fig. 15).

The successive process of prediction/experimental-data confrontation allows the optimal determination of the evolution laws of the cracks density, of the cracks thickness aspect ratio (Fig. 16) and of the volume concentration of the three cracks arrays (Fig. 17) [24].

Fig. 18 plots the experimental changes of the nine stiffnesses identified from the phase velocities as a function of tensile stress and the prediction, Eq. (17), plotted in straight lines.

Confrontation of predictions and experimental data has also been done for the strains in this composite. Experimentally, an extensometer gives us the total strain (Fig. 19). The elastic strain is computed from the elasticity tensor (Fig. 18). The difference between the elastic strain and the total strain is the inelastic strain (Fig. 19). Prediction of the effective elastic properties gives us the predicted elastic strain, Eq. (1). The inelastic

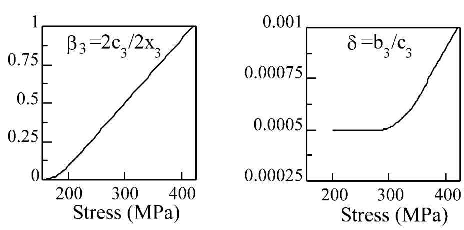

Fig. 10. Variation of the crack density parameter and of the thickness crack aspect ratio for the transverse matrix microcracking as a function of applied stress.

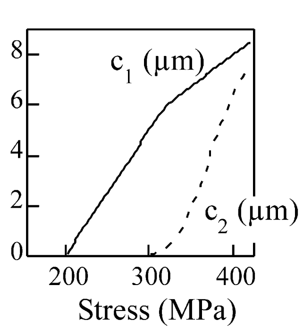

Fig. 11. Variation of the semi-axes of the growing longitudinal cracks as a function of applied stress.

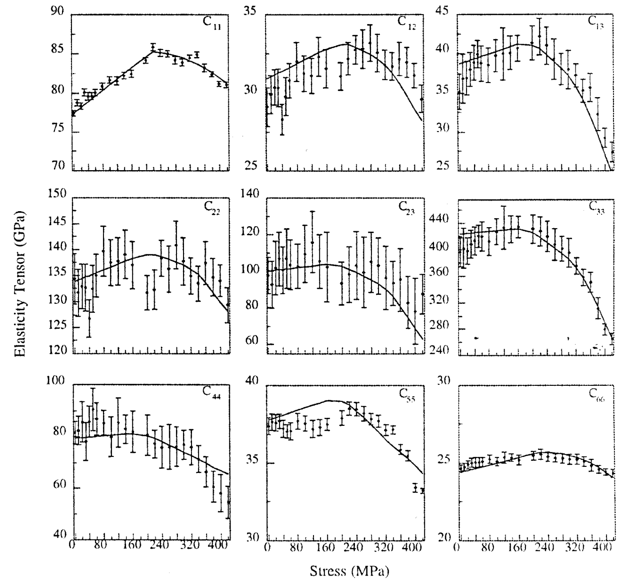

Fig. 12. Stiffness changes (in GPa) during a tensile test in fibres direction 3, 1D SiC–SiC.

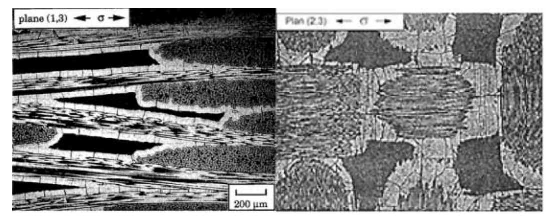

Fig. 13. Micrographs of the three cracks arrays in a 2D C–SiC.

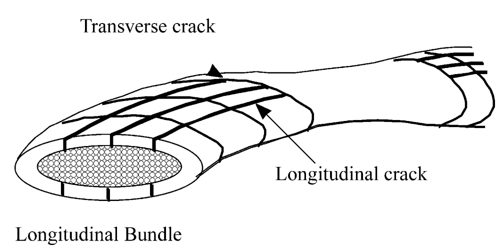

Fig. 14. Schematic description of fthe three cracks arrays in a longitudinal bundle of a 2D C–SiC.

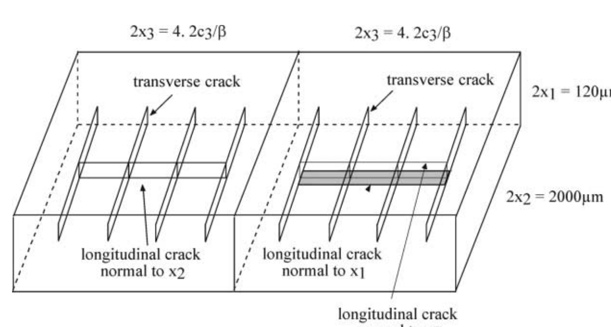

Fig. 15. Unit cell of the cracked 2D C–SiC.

strain is simply the effective density of transverse matrix microcracks multiplied by their aspect ratio, Eq. (4). Then, the predicted total strain is the sum of the predicted elastic and inelastic parts of the strain. The comparison between the experimental strains and their predictions (Fig. 19), allows us to validate the choice of the evolution laws of the two internal variables, β and δ, and, so, of the description of this micro level damage mechanism [24].

5. Cyclic loading

When cyclic loading is performed, as the stress decreases during unloading, the transverse cracks that have been created during the monotonic loading are prone to close (Fig. 20). They are still present anyway but their closure simulates an increase of stiffness as if the material could recover its mechanical properties. The cracks that remain open represent the active part of the microcracking. An apparent state of damage seems to decrease while the damage accumulated is greater.

The closing effect induces damage deactivation leading to unilateral behaviour [26–28]. The internal variable β is then representative of a damage that can be qualified as apparent damage [14]. It is necessary to make a difference between the apparent state of cracking and the number of cracks that has been created effectively. Just as the active cracks were defined, we can define an active or apparent crack density:

$$\beta^* = F{\cdot}\beta \tag{19}$$

where β is the density of cracks that have been accumulated during monotonic loading. The opening-closure variable F is the proportion of cracks that remains open during unloading and thus represents the damage deactivation.

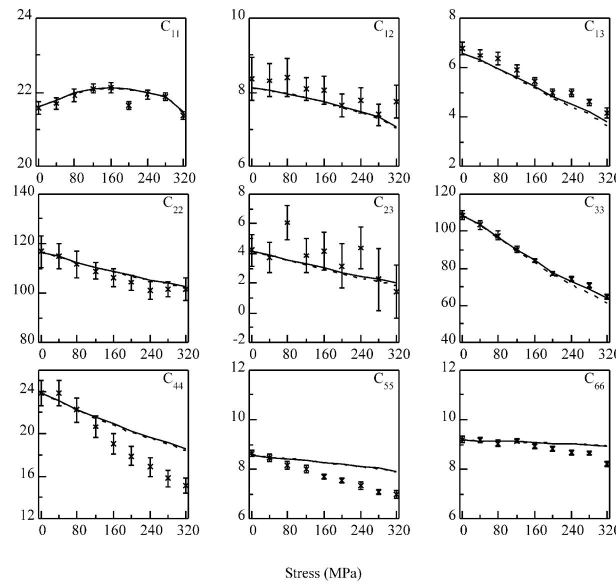

Fig. 18. Stiffness changes (in GPa) during a tensile test in fibres direction 3, 2D C–SiC.

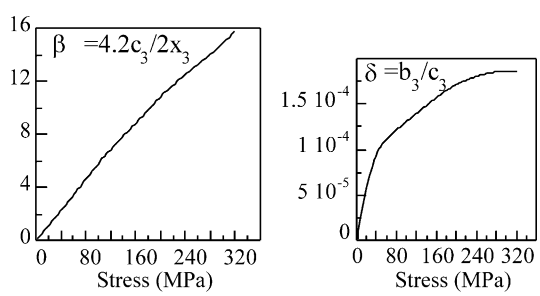

Fig. 16. Variation of the crack density parameter and of the thickness crack aspect ratio for the transverse matrix microcracking as a function of applied stress.

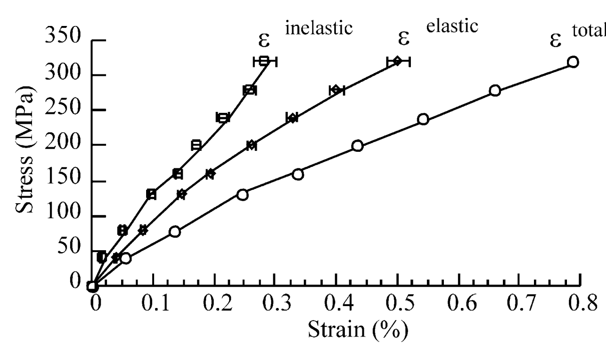

Fig. 19. Variation of the total strain, of the elastic strain and of the inelastic strain as a function of the applied stress.

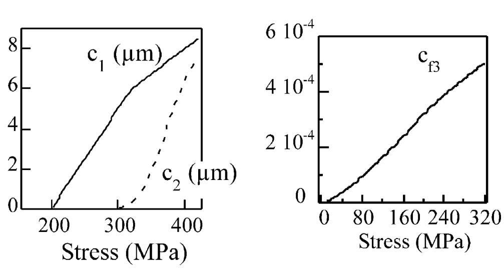

Fig. 17. Variation of the volume concentration of the three cracks arrays.

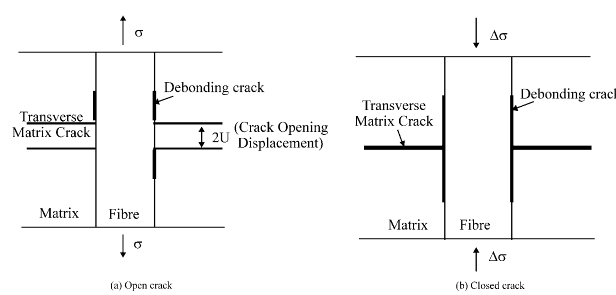

Fig. 20. States of the transverse cracks.

Under cyclic tensile loading, a 2D carbon SiC composite exhibits quite immediately a non-linear behaviour (Fig. 21). The cycles do not show any hysteresis but nevertheless the residual strains are far from negligible. The lack of hysteresis let think that the sliding occurs with little or no friction at the fibre matrix interface [15].

S_{33}, S_{44} and S_{55} modified by the presence of a transverse cracks system are the only compliances to exhibit a variation during the cycles (Fig. 22). Their progressive decrease during the unloading cycles is relevant of a progressive closure of the cracks. As some cracks close, they are active no more and they lose the effect they had on the compliances. When the sample is reloaded, all the cracks that have been created re open and become active again. The compliances reach the value they had before unloading [15].

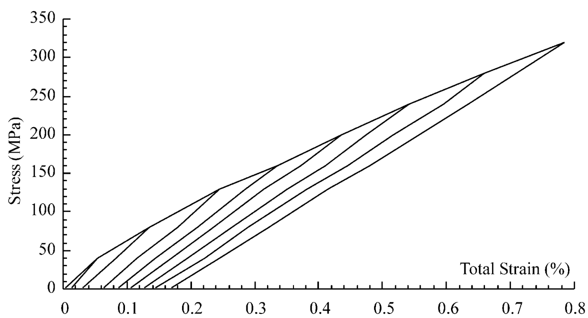

Fig. 21. Stress-strain curve of a 2D C–SiC under cyclic loading.

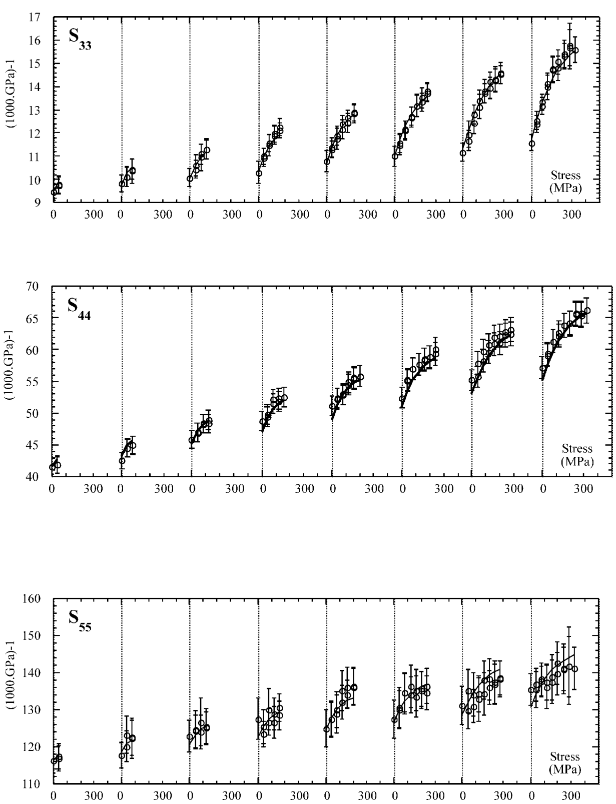

Fig. 22. The most influenced compliances of a 2D C–SiC under cyclic loading.

Fig. 23 shows the relative part of the elastic and inelastic strains on the total strain. As an example, the cycle at 280 MPa has been chosen. The linear variation of the total strain is the sum of both the non-linear variations of the elastic and inelastic strains. The elastic strain naturally comes back to zero with the stress whereas the inelastic strain is still present as a function of the transverse cracks which is still active [15].

The successive process of prediction experimental data confrontation allows us the optimal determination of the evolution laws of the cracks density and the opening closure variable (Fig. 24). Fig. 22 plots the experimental changes of the three most influenced compliances under cyclic loading and the prediction plotted in straight lines. According to Eqs. (17) and (19), the behaviour during the cycles is fully described by the variables: β, δ and F that predict the compliances variations (Fig. 22). During the loading/unloading cycles, only the transverse crack system has an influence on the behaviour due to the opening-closure of the cracks [15].

During unloading, it is obvious that the effect of closure of the cracks must be taken into account. The proportion of transverse cracks which is still active under cyclic loading is described with the opening-closure function F. Eq. (4) becomes:

$$\varepsilon^{\text{in}} = \delta \cdot F \cdot \beta \tag{20}$$

Furthermore, some cracks will probably not completely close and have a residual opening. This opening can be due to the roughness of the cracks edges coming from both wear debris at the sliding interface [29] and

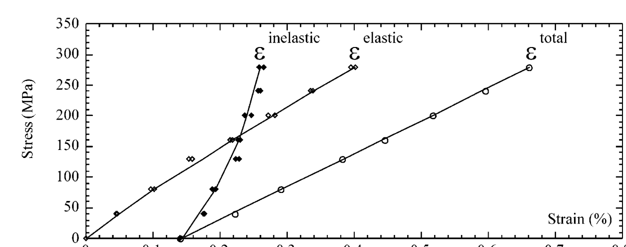

Fig. 23. Under load strain partition during cyclic loading at 280 MPa of a 2D C–SiC.

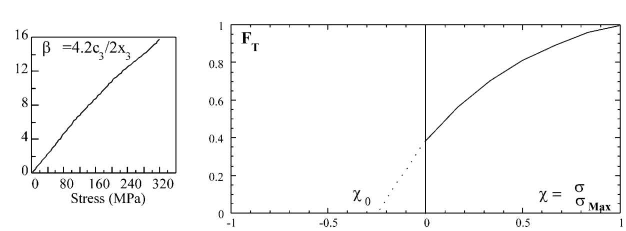

Fig. 24. Variation of the crack density parameter and of the opening-closure function of a 2D C–SiC.

wedging of the cracks by grain bridging [30]. To describe this later fact, another aspect ratio δ' that takes the residual opening into account is introduced [15]. Finally, the variations of inelastic strains during the whole test can be described by [15]:

$$\varepsilon^{\mathrm{in}} = \beta(F(\delta - \delta') + \delta') \tag{21}$$

The inelastic strains are thus a function of the transverse crack density and of the aspect ratio of the cracks whether they are completely open or not (Fig. 25).

The predictions of the three dimensional changes in elasticity and of the inelastic strains under cyclic loading are shown to compare favourably with experimental data. While the crack density describes the inelastic strains and the drop in elastic modulus, the opening closure variable modulates these effects when cyclic loading is applied [15].

6. Effective elastic stiffnesses of an anisotropic medium permeated by tilted cracks

In composite materials, failure mechanisms favour the generation of microcracks oriented normally to the tensile stress [1]. An off-axis tensile loading creates microcracks whose orientation does not coincide with the fibre axes [31,32] and induces a fully anisotropic elastic degradation. To emphasise the induced anisotropy and the loss of elastic symmetry caused by off-principal solicitations, the measurement of the changes of all the stiffness tensor components has been done [33] for a 2D C–C–SiC composite material, subjected to a tensile solicitation at 30° from one of the fibre directions (Fig. 26). The load-induced changes of the thirteen stiffnesses, associated with a monoclinic symmetry, have been recovered from ultrasonic velocity data [33].

Damage-induced changes of the thirteen stiffnesses (C_{ij}) identified in the geometric coordinate system of the sample are plotted in Fig. 27 as a function of the applied tensile stress [33]. The loss of stiffness along the tensile axis x_3 is very important and occurs from the first stress levels. The variation of the experimental stiffnesses C_{33}, C_{44} and C_{55} leads us to consider that the matrix microcracking oriented normally to the tensile loading is the main damage mode and induces a fully anisotropic degradation. In particular, the coupling stiffnesses C_{34}, C_{24}, C_{14} and C_{56}, which are naturally about zero at 0 MPa, become non-negligible at 40 MPa. After 60 MPa, they decrease and recover their initial value of about zero. It coincides with the increase of inelastic strain (Fig. 29).

The fibrous reinforcement stops and deviates the matrix microcracking in mode II. Sliding occurs in the fibre-matrix interphase and leads to other cracking modes consisting in slit cracks whose orientations coincide with

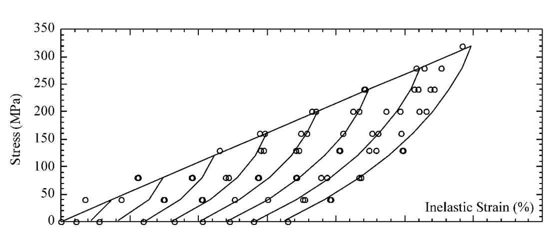

Fig. 25. Variation of the inelastic strains during cyclic loading of a 2D C–SiC.

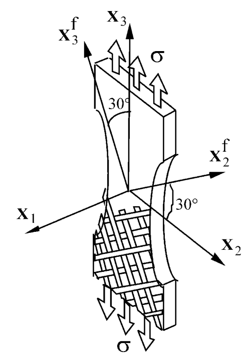

Fig. 26. The 30° off-axis solicitation sample, cut out according to a 30° angle from fibre axes, and loading in this direction.

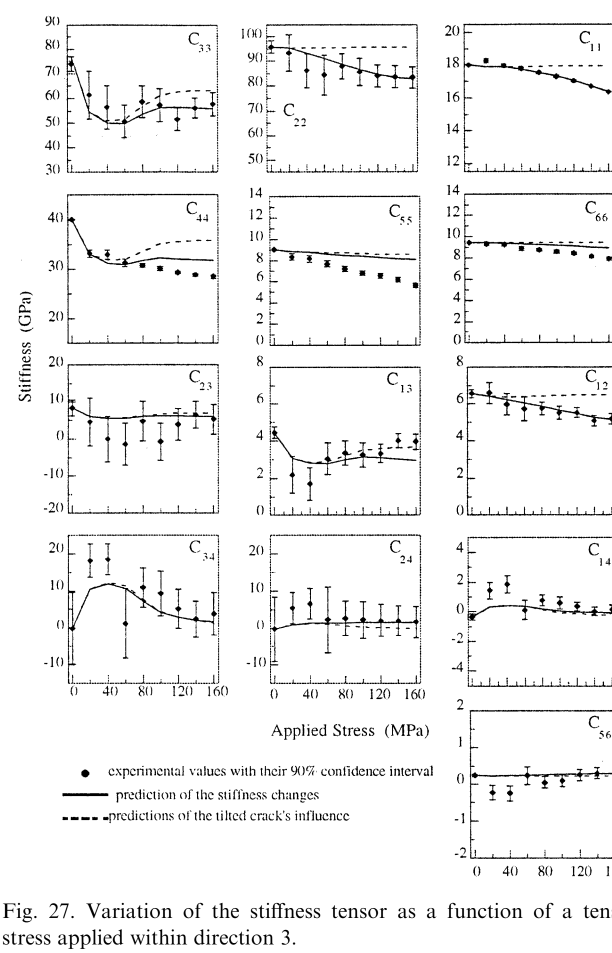

Fig. 27. Variation of the stiffness tensor as a function of a tensile stress applied within direction 3.

the directions of the fibres (Fig. 28).[34] The experimental changes of the stiffnesses C_{11}, C_{22}, C_{66} and C_{55} point out the generation of these cracking modes [33].

The under load strain partitio [8] in the direction x_3 with respect to the applied stress is plotted in Fig. 29. The part of inelastic strain increases with the stress. As it is the macroscopic result of the crack opening displacement [12], the inelastic strain is associated with an increase in the thickness of the matrix microcracks.

To describe the damage on the 2D C/C–SiC by establishing the relationship between the effective stiffness tensor and the intensity of damage, the matrix microcracking oriented normally to the tensile stress requires particular attention to fully understand the experimental changes observed during the load [34]. The material microstructure and the increase of inelastic strain lead us to consider 3D-defined tilted cracks for the evaluation of effective stiffness changes due to this cracking mode.

The tensile solicitation along a non-principal direction creates a matrix microcracking normal to the loading direction. The extension of the cracks is limited by the waviness of the bundles and experimental observation of inelastic strains implies that crack opening displacement is not negligible. Therefore, it is necessary to consider 3D-defined cracks and to calculate the *P*-tensor for 3D-defined tilted cracks in order to evaluate the effective stiffness tensor of the damaged material. As the classical formula of the *P*-tensor components is given in the crack's frame, Eq. (10), we have to consider a more general symmetry for the solid. Indeed, the material is no longer orthotropic but monoclinic when tilted cracks are created. Therefore, the prediction of the effective stiffness tensor requires the evaluation of 13 P_{ijkl} in order to study the damage process [34].

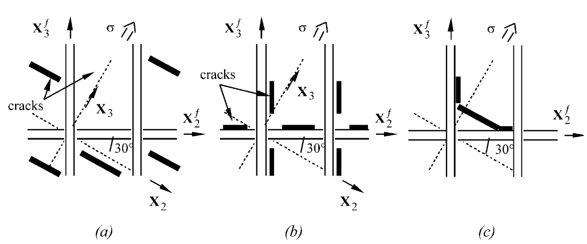

Fig. 28. Damage modes taking place during the 30° off-axis solicitation: (*a*) matrix microcracking normal to the loading direction, (*b*) deviation at the fibre-matrix interphase, (*c*) "zigzag" shape of the resulting microcracking.

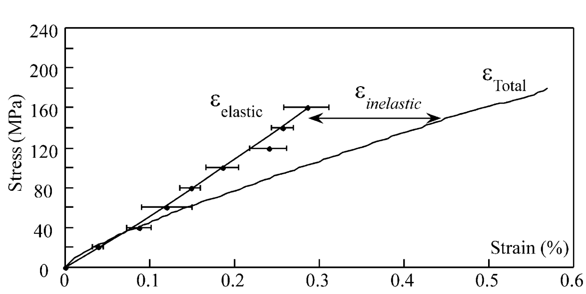

Fig. 29. Variation of the total strain, of the elastic strain and of the inelastic strain as a function of applied stress.

In the 2D C/C–SiC, the damage starts with the creation of cracks oriented normally to the stress direction (Fig. 30). As the governing equations lie to the effective stiffness tensor in the axes of the ellipsoidal crack, we have to consider the components of C in the principal coordinate system $R=(x_1, x_2, x_3)$ of the ellipsoid.

The 2D C/C–SiC is orthotropic in the material principal coordinate system $R^f=(x_1, x_2^f, x_3^f)$:

$$C_{ij}(R^f) = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ & C_{22} & C_{23} & 0 & 0 & 0 \\ & & C_{33} & 0 & 0 & 0 \\ & \text{Sym} & & C_{44} & 0 & 0 \\ & & & & C_{55} & 0 \\ & & & & & C_{66} \end{bmatrix}. \tag{22}$$

The stiffness tensor becomes in the coordinate system $R=(x_1, x_2, x_3)$, by using the fourth rank tensor rotation law,

$$(C_{ij}^R) = (M_{ij}^{R^f \to R})(C_{ij}^{R^f})(M_{ij}^{R^f \to R})^t, \tag{23}$$

where M is the transformation matrix:

$$(C_{ij}^R) = \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & 0 & 0 \\ & C_{22} & C_{23} & C_{24} & 0 & 0 \\ & & C_{33} & C_{34} & 0 & 0 \\ & \text{Sym} & & C_{44} & 0 & 0 \\ & & & & C_{55} & C_{56} \\ & & & & & C_{66} \end{bmatrix}. \tag{24}$$

As the cracks principal axes coincide with the coordinate system R, the material remains monoclinic during

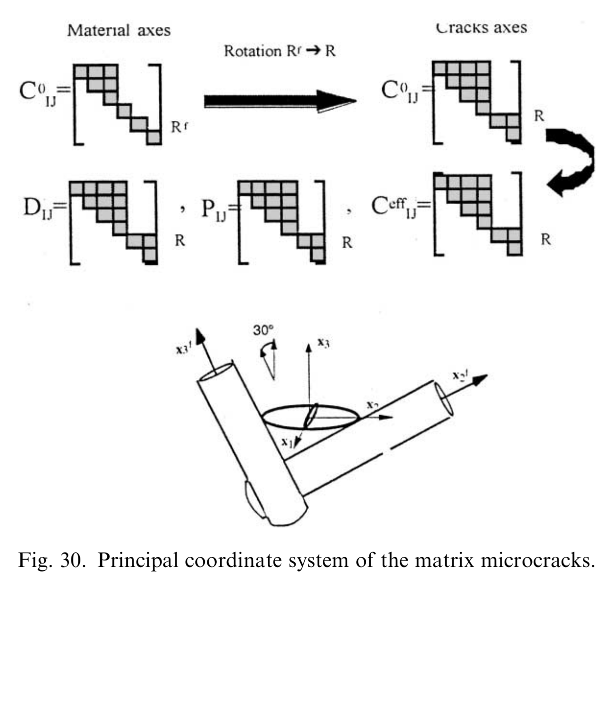

Fig. 30. Principal coordinate system of the matrix microcracks.

the damage process. The 3×3 matrix g and the fourth-order tensor D, Eq. (11), are calculated with the stiffness tensor given in Eq. (24) and D has the same form as C^R. It follows from Eq. (10) that the P and Q tensors have the same non-zero components as D and C^R [34].

6.1. Influence of the thickness of the crack

The experimental inelastic strain (Fig. 29) leads us to consider that a crack opening displacement occurs during the load for the tilted cracks (Fig. 31). To fully understand the relationship between the macroscopic changes and the microscopic phenomena, it is necessary to study the influence of the COD on the effective stiffness tensor [34].

Fig. 32 represents the variations of Q_{ij}^{-1} rather than C to insist on the influence of the thickness crack aspect ratio, $\delta = b/c$. The Q^{-1} tensor for a crack is calculated for an ellipticity crack aspect ratio, $\gamma = a/c$, equal to 0.1. Indeed in such a material, the waviness of the bundles limits the extension of the cracks. The results (Fig. 32) are given with respect to δ. We have plotted only the most affected components Q_{33}^{-1}, Q_{44}^{-1}, Q_{34}^{-1}, Q_{55}^{-1}, Q_{14}^{-1} and Q_{24}^{-1}.

The weaker δ is, the more important its influence on the variations of the considered components of the tensor Q^{-1} is. Whereas the three most affected components, Q_{33}^{-1}, Q_{44}^{-1} and Q_{34}^{-1} reach a similar rate for the weakest crack thickness aspect ratios, their variations are strongly different for higher crack thickness aspect ratios. Indeed, the value of Q_{34}^{-1} remains near zero up to $\delta = 0.001$ and then increases rapidly while Q_{33}^{-1} and Q_{44}^{-1} grow more regularly. Those variations throw light on the experimental results. At the beginning of load, the inelastic strain is about zero (Fig. 29) and the change of the coupling stiffnesses C_{34}, C_{24} and C_{14} is non negligible (Fig. 27). Then, after 60 MPa, as the part of inelastic strain increases, the COD starts and these stiffnesses decrease.

6.2. Stiffness changes due to the tilted cracks

The evaluation of the stiffness changes for a material containing tilted cracks is performed by using Eq. (17).

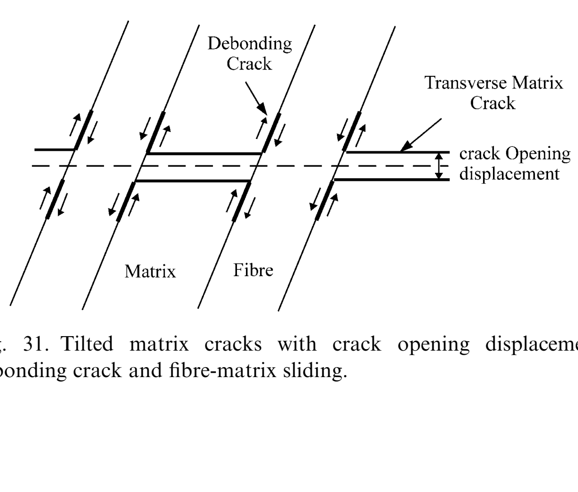

Fig. 31. Tilted matrix cracks with crack opening displacement, debonding crack and fibre-matrix sliding.

It requires the determination of the volume concentration of the cracks during the load. It is given by:

$$c_{fi} = \frac{4\pi}{3} \cdot \frac{a}{2x_1} \cdot \frac{b}{2x_2} \cdot \frac{b}{2x_3} = \frac{\pi}{6} \cdot \frac{2a}{2x_1} \cdot \frac{2c}{2x_2} \cdot \beta \cdot \delta, \tag{25}$$

where β is the crack density parameter in the direction 3, δ is the thickness crack aspect ratio, $2x_i$ is the distance between two cracks in the direction i and a, c and b are the semi-axes of the ellipsoid. As the waviness of the material limits the crack length in the plane (1, 2), $2a$, $2x_1$, $2c$ and $2x_2$ are constant and respectively equal to 150, 200, 1500 and 2000 μm [25]. Therefore, we have to identify the evolution laws of β and δ in order to evaluate the effective stiffness tensor C, by resolving the inverse problem [24]. Indeed, the experimental data, the stiffness changes and the inelastic strain, are used to compare the predictions obtained for different values of β and δ and allow the determination of the optimum variations of these two parameters (Fig. 33). The variations of the experimental stiffnesses C_{33}, C_{44} and C_{34} lead us to consider that the matrix microcracking starts with a weak thickness crack aspect ratio and a rapid increase of the density of cracks. Then after 60 MPa as the coupling stiffnesses C_{34}, C_{24} and C_{14} decrease more quickly than the others (Fig. 27), the crack opening displacement starts and the evolution of β is less important (Fig. 33).

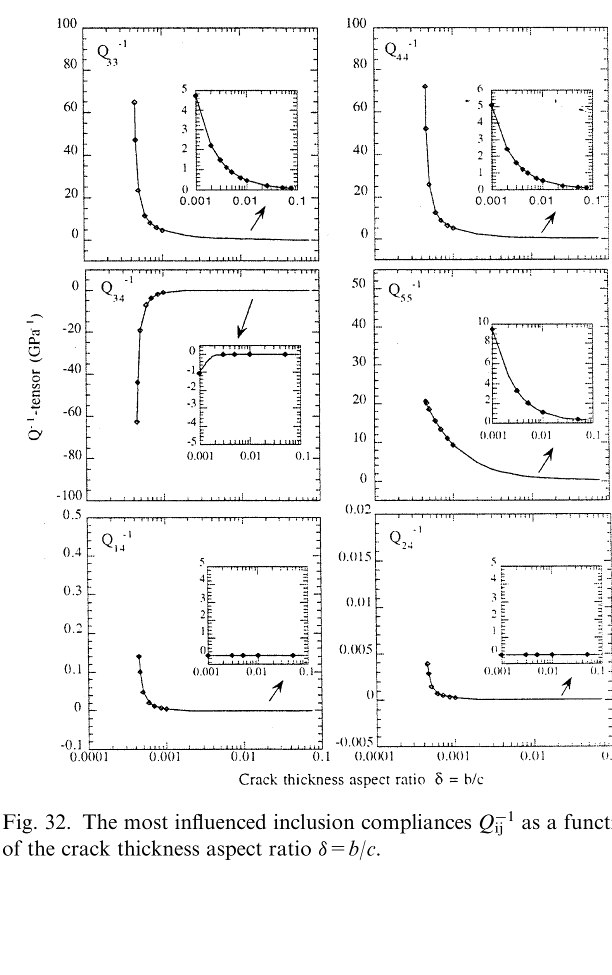

Fig. 32. The most influenced inclusion compliances Q_{ij}^{-1} as a function of the crack thickness aspect ratio $\delta = b/c$.

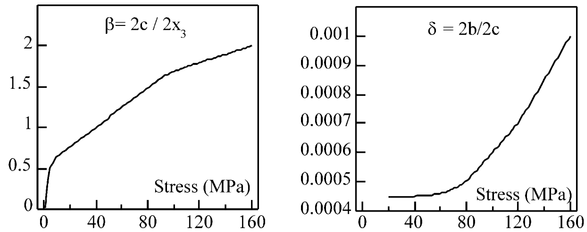

Fig. 33. Variation of the crack density parameter and of the thickness crack aspect ratio for the matrix microcracking normal to the loading direction as a function of applied stress.

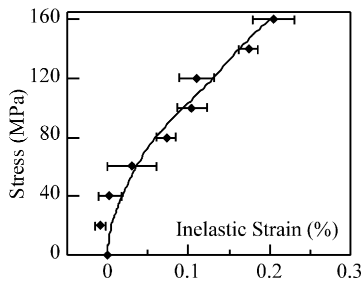

Fig. 34. Variation of the inelastic strain as a function of applied tensile stress.

Moreover, the comparison between the experimental inelastic strain [8] and its prediction (Fig. 34), given by Eq. (4), allows us to validate the choice of β and δ. There is a good agreement between experimental values of the inelastic strain and the predictions plotted in straight lines.

The effective stiffnesses are plotted in dotted line in Fig. 27. The loss of stiffness observed for C_{33} and C_{44} and the rapid increase of C_{34} at the beginning of the load is predicted. Moreover, the crack opening displacement taken into account after 60 MPa allows to recover the variation of the coupling stiffnesses C_{34}, C_{24} and C_{14} at the end of the damage process. It points out the influence of the thickness of the tilted cracks on the damage-induced anisotropy. Indeed, whereas cracks with weak thickness have a high anisotropic effect, the crack opening displacement induces a less important anisotropy [34].

7. Conclusion

The formulation of constitutive equations by using a mixed approach including a fully identification of the anisotropic damage and a taking into account of the whole strain and damage mechanisms, allows us to relate the micro- and macro-level damage measurement and provide coherent and comprehensive physical explanations for the observed experimental phenomenology. The major point is the non-arbitrariness of the choice of the internal variables with a definite physical meaning. The load-induced changes of the stiffness tensor can be described by a single parameter, the density of cracks. Moreover, taking into account the thickness of the cracks, i.e. the crack opening displacement, allows us to understand and to predict the experimental results.

Therefore the response of a damaged elastic material to any stress is totally predicted when the evolution of the crack density parameter and of the thickness crack aspect ratio is identified according to the crack orientation and the stress history.

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